



Cloud Computing

Assignment-Week 5

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

A cloud provider guarantees 99% availability. A customer's application operates 12 hours per day. What is the maximum allowable downtime during a 30-day month without violating the SLA?

- A. 3.6 hours
- B. 6 hours
- C. 7.2 hours
- D. 9 hours

Correct Answer: A

Detailed Solution: Scheduled time $12 \times 30 = 360$ hours

Allowed downtime : $1\% = 0.01 \times 360 = 3.6$ hours

QUESTION 2:

A cloud server dynamically activates additional CPU cores to handle a sudden spike in user requests. What is the most likely impact on the server's power consumption?

- A. Power consumption decreases because the workload is distributed.
- B. Power consumption remains unchanged since CPU frequency is fixed.
- C. Power consumption increases as additional CPU cores become active.
- D. Power consumption depends only on the amount of RAM used.

Correct Answer: C

Detailed Solution: Activating more CPU cores increases overall power consumption because each active core consumes power. Although techniques such as Dynamic Voltage and Frequency Scaling (DVFS) can improve energy efficiency, the total power generally increases as more cores are utilized.



QUESTION 3:

Which cloud characteristic most directly helps organizations avoid large upfront hardware investments?

- A. Elastic scalability
- B. Pay-as-you-go pricing
- C. Fault tolerance
- D. Multi-tenancy

Correct Answer: B

Detailed Solution: Pay-as-you-go pricing allows customers to pay only for consumed resources, eliminating the need for significant upfront hardware purchases.

QUESTION 4:

An HDFS cluster has a block size of 128 MB. A 700 MB input file is processed using MapReduce. Assuming one map task per HDFS block, how many map tasks are created?

- A. 5
- B. 6
- C. 7
- D. 8

Correct Answer: B

Detailed Solution: Blocks $700 \text{ MB} = 128 + 128 + 128 + 128 + 128 + 60 = 6 \text{ blocks}$ One mapper per block so 6 map tasks

QUESTION 5:

Which virtualization component is primarily responsible for creating and managing virtual machines?

- A. Operating System Kernel
- B. Hypervisor
- C. BIOS
- D. Device Driver

Correct Answer: B

Detailed Solution: A hypervisor (Virtual Machine Monitor) creates, runs, and manages virtual machines by abstracting the underlying physical hardware.



QUESTION 6:

Which MapReduce component determines which Reducer receives a particular key?

- A. Combiner
- B. Partitioner
- C. RecordWriter
- D. InputSplit

Correct Answer: B

Detailed Solution: The Partitioner determines the destination Reducer for each intermediate key. By default, Hadoop uses a hash-based partitioner to distribute keys among Reducers.

QUESTION 7:

Which MapReduce component is optional and may execute zero, one, or multiple times?

- A. Mapper
- B. Reducer
- C. Combiner
- D. RecordReader

Correct Answer: C

Detailed Solution: The **Combiner** is an optional optimization. Hadoop does **not guarantee** that it will execute, so algorithms must remain correct even if the Combiner is skipped.

QUESTION 8:

A cloud provider observes that many physical servers are running at only 15% CPU utilization. To reduce electricity costs, it migrates virtual machines onto fewer servers and powers off the idle hosts while ensuring that application performance remains within SLA limits.

Which scheduling strategy is being applied?

- A. Round-Robin Scheduling
- B. Energy-Aware Scheduling
- C. First-Come-First-Serve Scheduling
- D. Fair Scheduling

Correct Answer: B



Detailed Solution: Energy-aware scheduling consolidates workloads onto fewer physical machines, allowing unused servers to be switched to low-power or off states. This reduces energy consumption while maintaining the required Quality of Service (QoS).

QUESTION 9:

Statement 1: Elasticity enables cloud users to increase or decrease resources according to workload demand.

Statement 2: Elasticity helps organizations avoid paying for unused computing resources.

- A. Both statements are true.
- B. Both statements are false.
- C. Statement 1 is true and Statement 2 is false.
- D. Statement 1 is false and Statement 2 is true.

Correct Answer: A

Detailed Solution: Elasticity allows resources to scale dynamically. Since resources can be released when demand decreases, organizations avoid paying for idle infrastructure.

QUESTION 10:

Average demand = 75 units, Peak demand = 225 units, The cloud provider offers a utility premium of 2.8. Which statement is true?

- A. Cloud cost < Owning cost
- B. Cloud cost > Owning cost
- C. Both costs are equal
- D. None of the above

Correct Answer: A

Detailed Solution: Maximum premium $225/75=3$ Since $2.8 < 3$.Cloud remains cheaper.